

# Subspaces and Rank

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## Rank

**Definition 0.1** (Linearly independent vectors). Vectors  $\tilde{v}_1, \dots, \tilde{v}_k$  of the same length  $p$  are **linearly independent** if the only numbers  $c_1, \dots, c_k$  with

$$c_1 \tilde{v}_1 + \dots + c_k \tilde{v}_k = \tilde{0}$$

are

$$\begin{aligned} c_1 &= \dots \\ &= c_k \\ &= 0. \end{aligned}$$

Vectors that are not linearly independent are **linearly dependent**.

**Example 0.1** (Independent and dependent pairs of vectors).

- $\tilde{v}_1 = (1, 0)$  and  $\tilde{v}_2 = (1, 1)$  are linearly independent:  $c_1 \tilde{v}_1 + c_2 \tilde{v}_2 = (c_1 + c_2, c_2)$ , which is  $\tilde{0}$  only if  $c_2 = 0$  and then  $c_1 = 0$ .
- $\tilde{v}_1 = (1, 2)$  and  $\tilde{v}_2 = (2, 4)$  are not:  $2\tilde{v}_1 - \tilde{v}_2 = \tilde{0}$ .

**Definition 0.2** (Rank). The **rank** of a matrix  $\mathbf{A}$ , written  $\text{rank}(\mathbf{A})$ , is the largest number of columns of  $\mathbf{A}$  that are linearly independent (Definition 0.1).

**Example 0.2** (The rank of two  $3 \times 2$  matrices). The matrix  $\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$  has rank 2: if  $c_1(1, 1, 1) + c_2(1, 2, 3) = \tilde{0}$ , then subtracting the first entry from the second gives  $c_2 = 0$ ,

and then  $c_1 = 0$ .

The matrix  $\begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$  has rank 1: its second column is twice its first, so the two columns are not linearly independent, but the first column on its own is.

**Definition 0.3** (Full column rank). An  $n \times p$  matrix has **full column rank** if its rank (Definition 0.2) is  $p$ , so that all of its columns are linearly independent.

*Remark 0.1* (Full column rank needs at least as many rows as columns). In Example 0.2, the first matrix has full column rank: its rank is 2, and it has 2 columns. The second does not, because its rank is 1.

An  $n \times p$  matrix can have full column rank only when  $p \leq n$ , because more than  $n$  vectors of length  $n$  are never linearly independent. For example, the  $2 \times 3$  matrix  $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$  has 3 columns of length 2, and  $1 \cdot (1, 0) + 1 \cdot (0, 1) - 1 \cdot (1, 1) = \tilde{0}$ , with coefficients that are not all zero. So its 3 columns are not linearly independent, and its rank is less than 3.

## Subspaces

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**Definition 0.4** (Subspace (linear subspace, vector space, linear space)). A set  $\mathcal{S}$  of vectors in  $\mathbb{R}^p$  is a **subspace** of  $\mathbb{R}^p$  if it is not empty and it is closed under vector addition and scalar multiplication:

1. if  $\tilde{u} \in \mathcal{S}$  and  $\tilde{v} \in \mathcal{S}$ , then  $\tilde{u} + \tilde{v} \in \mathcal{S}$ ;
2. if  $\tilde{u} \in \mathcal{S}$  and  $c \in \mathbb{R}$ , then  $c\tilde{u} \in \mathcal{S}$ .

A subspace is also called a **linear subspace**, a **vector space**, or a **linear space**.

**Example 0.3** (A line through the origin is a subspace). Let  $\mathcal{S} = \{c(1, 2) : c \in \mathbb{R}\}$ , the line in  $\mathbb{R}^2$  through  $(0, 0)$  and  $(1, 2)$ .

- **Addition:**  $a(1, 2) + b(1, 2) = (a + b)(1, 2)$ , which is in  $\mathcal{S}$ .
- **Scalar multiplication:**  $c(a(1, 2)) = (ca)(1, 2)$ , which is in  $\mathcal{S}$ .

So  $\mathcal{S}$  is a subspace of  $\mathbb{R}^2$ . Two other subspaces of  $\mathbb{R}^p$  are the set  $\{\tilde{0}\}$  that holds only the zero vector, and  $\mathbb{R}^p$  itself.

**Example 0.4** (A line that misses the origin is not a subspace). Let  $\mathcal{T} = \{(x, 1) : x \in \mathbb{R}\}$ , the horizontal line in  $\mathbb{R}^2$  at height 1. The vectors  $(0, 1)$  and  $(2, 1)$  are in  $\mathcal{T}$ , but their sum  $(2, 2)$  is not, because its second entry is 2, not 1. So  $\mathcal{T}$  is not closed under addition, and it is not a subspace.

**Theorem 0.1** (Every subspace contains the zero vector). *If  $\mathcal{S}$  is a subspace of  $\mathbb{R}^p$  (Definition 0.4), then  $\tilde{0} \in \mathcal{S}$ .*

#### **i** Proof

*Proof.* A subspace is not empty, so it contains some vector  $\tilde{u}$ . Closure under scalar multiplication with  $c = 0$  puts  $0 \cdot \tilde{u}$  in  $\mathcal{S}$ , and  $0 \cdot \tilde{u} = \tilde{0}$ .  $\square$

**Example 0.5** (Using the zero vector to rule out a subspace). The plane  $\{(x, y, z) : x + y + z = 1\}$  in  $\mathbb{R}^3$  does not contain  $\tilde{0} = (0, 0, 0)$ , because  $0 + 0 + 0 = 0 \neq 1$ . So, by Theorem 0.1, it is not a subspace. The line  $\mathcal{T}$  in Example 0.4 fails the same test:

$(0, 0)$  is not in  $\mathcal{T}$ , because its second entry is not 1.

Containing  $\tilde{0}$  is necessary but not sufficient. The union of the two coordinate axes in  $\mathbb{R}^2$ ,  $\{(x, 0) : x \in \mathbb{R}\} \cup \{(0, y) : y \in \mathbb{R}\}$ , contains  $\tilde{0}$ , but  $(1, 0) + (0, 1) = (1, 1)$  lies on neither axis, so the union is not closed under addition and is not a subspace.

**Definition 0.5** (Span). The **span** of vectors  $\tilde{v}_1, \dots, \tilde{v}_k \in \mathbb{R}^p$  is the set of all their linear combinations:

$$\text{span}\{\tilde{v}_1, \dots, \tilde{v}_k\} \stackrel{\text{def}}{=} \{c_1 \tilde{v}_1 + \dots + c_k \tilde{v}_k : c_1, \dots, c_k \in \mathbb{R}\}.$$

The vectors  $\tilde{v}_1, \dots, \tilde{v}_k$  **span** a set  $\mathcal{S}$  if their span equals  $\mathcal{S}$ .

By convention, the span of the empty list ( $k = 0$ ) is  $\{\tilde{0}\}$ , and the empty list counts as linearly independent (Definition 0.1), since it has no coefficients that could fail to be zero.

**Example 0.6** (Two spans in  $\mathbb{R}^3$ ).

- $\text{span}\{(1, 0, 0), (0, 1, 0)\}$  is the set of vectors  $(c_1, c_2, 0)$ , the plane  $z = 0$  in  $\mathbb{R}^3$ .
- $\text{span}\{(1, 2, 3), (2, 4, 6)\}$  is only the line  $\{c(1, 2, 3) : c \in \mathbb{R}\}$ , because  $(2, 4, 6) = 2(1, 2, 3)$ , so  $c_1(1, 2, 3) + c_2(2, 4, 6) = (c_1 + 2c_2)(1, 2, 3)$ . Adding a vector to a list does not always enlarge its span.

**Theorem 0.2** (A span is a subspace). *For any vectors  $\tilde{v}_1, \dots, \tilde{v}_k \in \mathbb{R}^p$ ,  $\text{span}\{\tilde{v}_1, \dots, \tilde{v}_k\}$  (Definition 0.5) is a subspace of  $\mathbb{R}^p$  (Definition 0.4).*

### Proof

*Proof.* The span is not empty: it contains  $0 \cdot \tilde{v}_1 + \dots + 0 \cdot \tilde{v}_k = \tilde{0}$ .

**Addition.** Take two vectors in the span,  $\tilde{u} = a_1 \tilde{v}_1 + \dots + a_k \tilde{v}_k$  and  $\tilde{w} = b_1 \tilde{v}_1 + \dots + b_k \tilde{v}_k$ . Then

$$\begin{aligned} \tilde{u} + \tilde{w} &= (a_1 \tilde{v}_1 + \dots + a_k \tilde{v}_k) + (b_1 \tilde{v}_1 + \dots + b_k \tilde{v}_k) && \text{(substitute the two combinations)} \\ &= (a_1 + b_1) \tilde{v}_1 + \dots + (a_k + b_k) \tilde{v}_k && \text{(regroup, and distribute each } \tilde{v}_i \text{)} \end{aligned}$$

which is a linear combination of  $\tilde{v}_1, \dots, \tilde{v}_k$ , so it is in the span.

**Scalar multiplication.** For  $c \in \mathbb{R}$ ,

$$\begin{aligned} c \tilde{u} &= c(a_1 \tilde{v}_1 + \dots + a_k \tilde{v}_k) && \text{(substitute the combination)} \\ &= (ca_1) \tilde{v}_1 + \dots + (ca_k) \tilde{v}_k && \text{(distribute } c \text{ over the sum)} \end{aligned}$$

which is again in the span. □

**Example 0.7** (The line in Example 0.3 is a span). The line  $\mathcal{S} = \{c(1, 2) : c \in \mathbb{R}\}$  in Example 0.3 is  $\text{span}\{(1, 2)\}$ , so Theorem 0.2 gives a second proof that it is a subspace. Likewise, the plane  $z = 0$  in Example 0.6 is a subspace of  $\mathbb{R}^3$ , because it is  $\text{span}\{(1, 0, 0), (0, 1, 0)\}$ .

**Theorem 0.3** (A matrix-vector product combines the columns). *If  $\mathbf{A}$  is an  $m \times n$  matrix with columns  $\tilde{a}_1, \dots, \tilde{a}_n$  and  $\tilde{x} \in \mathbb{R}^n$ , then*

$$\mathbf{A}\tilde{x} = x_1\tilde{a}_1 + \dots + x_n\tilde{a}_n.$$

### **i** Proof

*Proof.* Compare entry  $i$  of the two sides, for each  $i = 1, \dots, m$ . Entry  $i$  of  $\tilde{a}_j$  is  $a_{ij}$ , so

$$\begin{aligned} (\mathbf{A}\tilde{x})_i &= a_{i1}x_1 + \dots + a_{in}x_n && \text{(Definition 11 in Matrices)} \\ &= x_1a_{i1} + \dots + x_na_{in} && \text{(multiplication of numbers is commutative)} \\ &= (x_1\tilde{a}_1 + \dots + x_n\tilde{a}_n)_i && \text{(entry } i \text{ of each } x_j\tilde{a}_j \text{ is } x_ja_{ij}) \end{aligned}$$

□

**Example 0.8** (A product as a combination of columns).

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} &= 1 \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 2 \\ 4 \end{bmatrix} && \text{(Theorem 0.3)} \\ &= \begin{bmatrix} -1 \\ -1 \end{bmatrix}, && \text{(arithmetic)} \end{aligned}$$

the same answer as computing each entry as a row-times-vector dot product (Remark 3 in Matrices).

**Theorem 0.4** (A matrix maps a combination to the same combination of images). *If  $\mathbf{A}$  is an  $m \times n$  matrix,  $\tilde{z}_1, \dots, \tilde{z}_k \in \mathbb{R}^n$  and  $c_1, \dots, c_k \in \mathbb{R}$ , then*

$$\mathbf{A}(c_1\tilde{z}_1 + \dots + c_k\tilde{z}_k) = c_1\mathbf{A}\tilde{z}_1 + \dots + c_k\mathbf{A}\tilde{z}_k.$$

### **i** Proof

*Proof.* The function  $\tilde{x} \mapsto \mathbf{A}\tilde{x}$  is a linear map by Theorem 9 in Matrices, and a linear map preserves linear combinations by Theorem 10 in Matrices. Applying that theorem with  $f(\tilde{x}) = \mathbf{A}\tilde{x}$  and  $\tilde{v}_i = \tilde{z}_i$  gives the identity. □

**Example 0.9** (Applying a matrix to a combination). Let  $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ ,  $\tilde{z}_1 = (1, 0)$  and  $\tilde{z}_2 = (0, 1)$ . Then  $\mathbf{A}\tilde{z}_1 = (1, 3)$  and  $\mathbf{A}\tilde{z}_2 = (2, 4)$ , and for the combination  $2\tilde{z}_1 - \tilde{z}_2 = (2, -1)$ :

$$\begin{aligned} \mathbf{A}(2, -1) &= (1 \cdot 2 + 2 \cdot (-1), 3 \cdot 2 + 4 \cdot (-1)) && \text{(Definition 11 in Matrices)} \\ &= (0, 2) && \text{(arithmetic)} \\ &= 2(1, 3) - (2, 4), && \text{(arithmetic)} \end{aligned}$$

which is  $2\mathbf{A}\tilde{z}_1 - \mathbf{A}\tilde{z}_2$ , as Theorem 0.4 says.

**Theorem 0.5** (More than  $p$  vectors in  $\mathbb{R}^p$  are linearly dependent). *If  $k > p$ , then any  $k$  vectors  $\tilde{v}_1, \dots, \tilde{v}_k \in \mathbb{R}^p$  are not linearly independent (Definition 0.1): some numbers  $c_1, \dots, c_k$ , not all zero, satisfy  $c_1\tilde{v}_1 + \dots + c_k\tilde{v}_k = \tilde{0}_p$ .*

### Proof

*Proof.* **Induction** on  $p$ .

**Base case**,  $p = 1$ . Each  $\tilde{v}_i$  is a single number  $v_i$ , and  $k \geq 2$ . If  $v_1 = 0$ , take  $c_1 = 1$  and every other  $c_i = 0$ . Otherwise take  $c_1 = v_2$ ,  $c_2 = -v_1$  and every other  $c_i = 0$ ; then  $c_2 \neq 0$ , and  $v_2v_1 - v_1v_2 = 0$ .

**Inductive step.** Let  $p \geq 2$ , assume the **theorem** holds for vectors in  $\mathbb{R}^{p-1}$ , and take  $k > p$  vectors in  $\mathbb{R}^p$ . Write  $v_{i,p}$  for the last entry of  $\tilde{v}_i$ .

*Case 1: every  $v_{i,p} = 0$ .* Deleting the last entry gives  $k > p - 1$  vectors in  $\mathbb{R}^{p-1}$ , which by the inductive hypothesis have a combination with coefficients not all zero that equals  $\tilde{0}_{p-1}$ . The same coefficients combine the original vectors to  $\tilde{0}_p$ , since every last entry is 0.

*Case 2: some  $v_{i,p} \neq 0$ .* Renumber the vectors so that  $v_{k,p} \neq 0$ , and for  $i = 1, \dots, k - 1$  let

$$\tilde{w}_i \stackrel{\text{def}}{=} \tilde{v}_i - \frac{v_{i,p}}{v_{k,p}} \tilde{v}_k.$$

The last entry of  $\tilde{w}_i$  is  $v_{i,p} - \frac{v_{i,p}}{v_{k,p}} v_{k,p} = 0$ . Deleting that zero last entry gives  $k - 1 > p - 1$  vectors in  $\mathbb{R}^{p-1}$ , so by the inductive hypothesis there are numbers  $c_1, \dots, c_{k-1}$ , not all zero, whose combination of the shortened vectors is  $\tilde{0}_{p-1}$ . Putting the zero last entries back gives  $c_1\tilde{w}_1 + \dots + c_{k-1}\tilde{w}_{k-1} = \tilde{0}_p$ . Then

$$\begin{aligned}
\tilde{0}_p &= c_1 \tilde{w}_1 + \cdots + c_{k-1} \tilde{w}_{k-1} && \text{(choice of } c_1, \dots, c_{k-1} \text{)} \\
&= \sum_{i=1}^{k-1} c_i \left( \tilde{v}_i - \frac{v_{i,p}}{v_{k,p}} \tilde{v}_k \right) && \text{(definition of } \tilde{w}_i \text{)} \\
&= \sum_{i=1}^{k-1} c_i \tilde{v}_i - \sum_{i=1}^{k-1} \frac{c_i v_{i,p}}{v_{k,p}} \tilde{v}_k && \text{(distribute each } c_i, \text{ and split the sum)} \\
&= c_1 \tilde{v}_1 + \cdots + c_{k-1} \tilde{v}_{k-1} - \left( \sum_{i=1}^{k-1} \frac{c_i v_{i,p}}{v_{k,p}} \right) \tilde{v}_k, && \text{(factor } \tilde{v}_k \text{ out of the second sum)}
\end{aligned}$$

a combination of  $\tilde{v}_1, \dots, \tilde{v}_k$  equal to  $\tilde{0}_p$  whose first  $k-1$  coefficients are not all zero.  $\square$

**Example 0.10** (Three vectors in  $\mathbb{R}^2$ ). Theorem 0.5 says the three vectors  $(1, 0)$ ,  $(0, 1)$  and  $(1, 1)$  in  $\mathbb{R}^2$  are linearly dependent, and indeed  $1 \cdot (1, 0) + 1 \cdot (0, 1) - 1 \cdot (1, 1) = (0, 0)$ . The theorem says nothing about  $k \leq p$  vectors:  $(1, 0)$  and  $(0, 1)$  are linearly independent, while  $(1, 2)$  and  $(2, 4)$  are not (Example 0.1).

**Definition 0.6** (Basis). A **basis** of a subspace  $\mathcal{S}$  (Definition 0.4) is a list of vectors that are linearly independent (Definition 0.1) and span  $\mathcal{S}$  (Definition 0.5).

**Example 0.11** (Two bases of  $\mathbb{R}^2$ ).

- $(1, 0)$  and  $(0, 1)$  form a basis of  $\mathbb{R}^2$ : they are linearly independent, and any  $(x, y) \in \mathbb{R}^2$  equals  $x(1, 0) + y(0, 1)$ .
- $(1, 0)$  and  $(1, 1)$  also form a basis of  $\mathbb{R}^2$ : Example 0.1 shows they are linearly independent, and any  $(x, y)$  equals  $(x - y)(1, 0) + y(1, 1)$ , because  $(x - y) + y = x$  in the first entry and  $0 + y = y$  in the second.

**Example 0.12** (Lists that are not bases of  $\mathbb{R}^2$ ).

- $(1, 0)$  and  $(2, 0)$  are not a basis of  $\mathbb{R}^2$ . They are not linearly independent, since  $2(1, 0) - (2, 0) = \tilde{0}$ , and they do not span  $\mathbb{R}^2$ : every combination  $c_1(1, 0) + c_2(2, 0)$  has second entry 0, so  $(0, 1)$  is not in their span.
- $(1, 0)$ ,  $(0, 1)$  and  $(1, 1)$  span  $\mathbb{R}^2$ , but they are not a basis:  $(1, 0) + (0, 1) - (1, 1) = \tilde{0}$ , so they are not linearly independent.

**Theorem 0.6** (Coefficients in a basis are unique). *If  $\tilde{a}_1, \dots, \tilde{a}_k$  is a basis of a subspace  $\mathcal{S}$  (Definition 0.6), then every  $\tilde{x} \in \mathcal{S}$  is a linear combination  $\tilde{x} = c_1\tilde{a}_1 + \dots + c_k\tilde{a}_k$  for exactly one list of numbers  $c_1, \dots, c_k$ .*

### Proof

*Proof.* At least one such list exists, because the basis spans  $\mathcal{S}$  (Definition 0.5). Suppose two lists both work:

$$\begin{aligned} c_1\tilde{a}_1 + \dots + c_k\tilde{a}_k &= \tilde{x} \\ &= d_1\tilde{a}_1 + \dots + d_k\tilde{a}_k. \end{aligned}$$

Subtracting the right side from the left side gives

$$\begin{aligned} \tilde{0} &= (c_1\tilde{a}_1 + \dots + c_k\tilde{a}_k) - (d_1\tilde{a}_1 + \dots + d_k\tilde{a}_k) \quad (\text{both combinations equal } \tilde{x}) \\ &= (c_1 - d_1)\tilde{a}_1 + \dots + (c_k - d_k)\tilde{a}_k \quad (\text{regroup, and distribute each } \tilde{a}_i) \end{aligned}$$

Because  $\tilde{a}_1, \dots, \tilde{a}_k$  are linearly independent (Definition 0.1), every coefficient  $c_i - d_i$  is 0, so  $c_i = d_i$  for each  $i$ .  $\square$

**Example 0.13** (Coefficients of  $(3, 5)$  in two bases). In the basis  $(1, 0), (0, 1)$  of Example 0.11,  $(3, 5) = 3(1, 0) + 5(0, 1)$ , and no other coefficients work. In the basis  $(1, 0), (1, 1)$ , the formula in Example 0.11 gives

$$\begin{aligned} (3, 5) &= (3 - 5)(1, 0) + 5(1, 1) \\ &= -2(1, 0) + 5(1, 1). \end{aligned}$$

The same vector has different coefficients in different bases, but within one basis its coefficients are unique.

The list  $(1, 0), (0, 1), (1, 1)$  in Example 0.12 spans  $\mathbb{R}^2$  but is not linearly independent, and the coefficients of  $(3, 5)$  are not unique:

$$\begin{aligned} (3, 5) &= 3(1, 0) + 5(0, 1) + 0(1, 1) \\ &= 0(1, 0) + 2(0, 1) + 3(1, 1). \end{aligned}$$

**Definition 0.7** (Coordinates in a basis). Let  $\tilde{a}_1, \dots, \tilde{a}_k$  be a basis of a subspace  $\mathcal{S}$  (Definition 0.6), and let  $\tilde{x} \in \mathcal{S}$ . The **coordinates** of  $\tilde{x}$  in this basis are the numbers  $c_1, \dots, c_k$  with  $\tilde{x} = c_1\tilde{a}_1 + \dots + c_k\tilde{a}_k$ , and  $(c_1, \dots, c_k) \in \mathbb{R}^k$  is the **coordinate vector** of  $\tilde{x}$ . By Theorem 0.6 there is exactly one such list of numbers.

**Example 0.14** (Coordinates of  $(4, 1)$  in two bases of  $\mathbb{R}^2$ ). In the basis  $(1, 0), (1, 1)$  of Example 0.11, the coordinates of  $(4, 1)$  are 3 and 1:

$$\begin{aligned} 3(1, 0) + 1(1, 1) &= (3 + 1, 0 + 1) \quad (\text{Definition 8 in Vectors, entry by entry}) \\ &= (4, 1). \quad (\text{add}) \end{aligned}$$

In the basis  $(1, 0), (0, 1)$ , its coordinates are its entries, 4 and 1.

**Theorem 0.7** (All bases of a subspace have the same number of vectors). *If  $\tilde{a}_1, \dots, \tilde{a}_k$  and  $\tilde{b}_1, \dots, \tilde{b}_l$  are both bases of a subspace  $\mathcal{S}$  of  $\mathbb{R}^p$  (Definition 0.6), then  $k = l$ .*

### **i** Proof

*Proof.* Let  $\mathbf{A}$  be the  $p \times k$  matrix with columns  $\tilde{a}_1, \dots, \tilde{a}_k$ , and  $\mathbf{B}$  the  $p \times l$  matrix with columns  $\tilde{b}_1, \dots, \tilde{b}_l$ .

**Write  $\mathbf{A}$  in terms of  $\mathbf{B}$ .** Each  $\tilde{a}_j$  is in  $\mathcal{S}$ , and the  $\tilde{b}$ 's span  $\mathcal{S}$  (Definition 0.5), so  $\tilde{a}_j = \mathbf{B}\tilde{c}_j$  for some  $\tilde{c}_j \in \mathbb{R}^l$  (Definition 11 in Matrices). Let  $\mathbf{C}$  be the  $l \times k$  matrix with columns  $\tilde{c}_1, \dots, \tilde{c}_k$ . Then  $\underset{p \times k}{\mathbf{A}} = \underset{p \times l}{\mathbf{B}} \underset{l \times k}{\mathbf{C}}$ , because column  $j$  of  $\mathbf{BC}$  is  $\mathbf{B}\tilde{c}_j$ .

**The columns of  $\mathbf{C}$  are linearly independent.** Suppose  $\mathbf{C}\tilde{x} = \tilde{0}$  for some  $\tilde{x} \in \mathbb{R}^k$ . Then

$$\begin{aligned} \mathbf{A}\tilde{x} &= (\mathbf{BC})\tilde{x} \quad (\text{substitute } \mathbf{A} = \mathbf{BC}) \\ &= \mathbf{B}(\mathbf{C}\tilde{x}) \quad (\text{Theorem 5 in Matrices}) \\ &= \mathbf{B}\tilde{0} \quad (\text{substitute } \mathbf{C}\tilde{x} = \tilde{0}) \\ &= \tilde{0} \quad (\text{a matrix times the zero vector is } \tilde{0}) \end{aligned}$$

$\mathbf{A}\tilde{x}$  is the combination  $x_1\tilde{a}_1 + \dots + x_k\tilde{a}_k$  (Theorem 0.3), and the  $\tilde{a}$ 's are linearly independent (Definition 0.1), so  $\tilde{x} = \tilde{0}$ . So the only combination of the columns of  $\mathbf{C}$  that equals  $\tilde{0}$  has all coefficients 0.

**Count.**  $\mathbf{C}$  has  $k$  linearly independent columns of length  $l$ . More than  $l$  vectors in  $\mathbb{R}^l$  are never linearly independent (Theorem 0.5), so  $k \leq l$ . Exchanging the roles of the two bases gives  $l \leq k$ , so  $k = l$ .  $\square$

**Example 0.15** (Both bases of  $\mathbb{R}^2$  in Example 0.11 have two vectors). The bases  $(1, 0), (0, 1)$  and  $(1, 0), (1, 1)$  of  $\mathbb{R}^2$  in Example 0.11 both have 2 vectors, as Theorem 0.7 requires. So no list of 3 vectors is a basis of  $\mathbb{R}^2$ , which agrees with Example 0.12: the list  $(1, 0), (0, 1), (1, 1)$  spans  $\mathbb{R}^2$  but is not linearly independent.

**Definition 0.8** (Dimension). The **dimension** of a subspace  $\mathcal{S}$ , written  $\dim(\mathcal{S})$ , is the number of vectors in any basis of  $\mathcal{S}$  (Definition 0.6). Theorem 0.7 makes this number well defined. By convention, the empty list is the basis of  $\{\tilde{0}\}$ , so  $\dim(\{\tilde{0}\}) = 0$ .

**Example 0.16** (The dimensions of some subspaces).

- $\dim(\{\tilde{0}\}) = 0$ , by the convention in Definition 0.8.
- $\dim(\mathbb{R}^2) = 2$ , by Example 0.11. More generally,  $\dim(\mathbb{R}^p) = p$ : the indicator vectors  $\tilde{e}_1, \dots, \tilde{e}_p$  (Definition 12 in Vectors) are linearly independent, since  $c_1\tilde{e}_1 + \dots + c_p\tilde{e}_p = (c_1, \dots, c_p)$ , which is  $\tilde{0}_p$  only if every  $c_i = 0$ ; and they span  $\mathbb{R}^p$ , since any  $\tilde{x} = x_1\tilde{e}_1 + \dots + x_p\tilde{e}_p$ .
- The line in Example 0.3 has dimension 1: the single vector  $(1, 2)$  is linearly independent and spans the line.
- The plane  $z = 0$  in Example 0.6 has dimension 2, with basis  $(1, 0, 0), (0, 1, 0)$ .

**Theorem 0.8** (Adding a vector outside the span keeps a list independent). *If  $\tilde{v}_1, \dots, \tilde{v}_k$  are linearly independent (Definition 0.1) and  $\tilde{w}$  is not in  $\text{span}\{\tilde{v}_1, \dots, \tilde{v}_k\}$  (Definition 0.5), then  $\tilde{v}_1, \dots, \tilde{v}_k, \tilde{w}$  are linearly independent.*

#### **i** Proof

*Proof.* Suppose  $c_1\tilde{v}_1 + \dots + c_k\tilde{v}_k + d\tilde{w} = \tilde{0}$ .

**The coefficient  $d$  is 0.** If  $d \neq 0$ , then

$$\begin{aligned} \tilde{w} &= \frac{1}{d}(d\tilde{w}) && (d \neq 0) \\ &= \frac{1}{d}(-c_1\tilde{v}_1 - \dots - c_k\tilde{v}_k) && (\text{solve the supposed equation for } d\tilde{w}) \\ &= \left(-\frac{c_1}{d}\right)\tilde{v}_1 + \dots + \left(-\frac{c_k}{d}\right)\tilde{v}_k, && (\text{distribute } \frac{1}{d}) \end{aligned}$$

which puts  $\tilde{w}$  in  $\text{span}\{\tilde{v}_1, \dots, \tilde{v}_k\}$ , contrary to the assumption.

**The other coefficients are 0.** With  $d = 0$ , the equation reads  $c_1\tilde{v}_1 + \dots + c_k\tilde{v}_k = \tilde{0}$ , so every  $c_i = 0$ , because  $\tilde{v}_1, \dots, \tilde{v}_k$  are linearly independent.  $\square$

**Example 0.17** (Adding  $(0, 0, 1)$  to two independent vectors in  $\mathbb{R}^3$ ). The vectors  $(1, 0, 0)$  and  $(0, 1, 0)$  are linearly independent, and their span is the plane  $z = 0$  (Example 0.6). The vector  $(0, 0, 1)$  has third entry 1, so it is not in that plane, and by Theorem 0.8 the three vectors  $(1, 0, 0), (0, 1, 0), (0, 0, 1)$  are linearly independent. Adding  $(1, 1, 0)$  instead would not work:  $(1, 1, 0)$  is in the span, and  $(1, 0, 0) + (0, 1, 0) - (1, 1, 0) = \tilde{0}$ .

**Theorem 0.9** (An independent list extends to a basis). *If  $\mathcal{S}$  is a subspace of  $\mathbb{R}^p$  (Definition 0.4) and  $\tilde{v}_1, \dots, \tilde{v}_k \in \mathcal{S}$  are linearly independent (Definition 0.1), then adding vectors of  $\mathcal{S}$  to the list gives a basis of  $\mathcal{S}$  (Definition 0.6). Starting from the empty list shows that every subspace has a basis.*

**i** Proof

*Proof.* Repeat the following step. If the current list spans  $\mathcal{S}$ , stop: it is linearly independent and spans  $\mathcal{S}$ , so it is a basis. Otherwise, some  $\tilde{w} \in \mathcal{S}$  is not in the span of the current list; add  $\tilde{w}$  to the list, which stays linearly independent by Theorem 0.8.

Every list produced is a linearly independent list of vectors in  $\mathbb{R}^p$ , so it has at most  $p$  vectors (Theorem 0.5). Each step adds one vector, so the process stops after at most  $p - k$  steps, and when it stops the list is a basis of  $\mathcal{S}$ .

For the empty list, which is linearly independent and spans  $\{\tilde{0}\}$  by the conventions in Definition 0.5, the first step asks whether  $\mathcal{S} = \{\tilde{0}\}$ : if so, the empty list is already a basis; if not, the process adds vectors as described.  $\square$

**Example 0.18** (Extending  $(1, 1, 0)$  to a basis of  $\mathbb{R}^3$ ). Start with the single vector  $(1, 1, 0)$ .

1.  $(1, 0, 0)$  is not in  $\text{span}\{(1, 1, 0)\}$ , because every multiple  $c(1, 1, 0)$  has equal first and second entries. Add it.
2.  $(0, 0, 1)$  is not in  $\text{span}\{(1, 1, 0), (1, 0, 0)\}$ , because every combination of those two vectors has third entry 0. Add it.
3. The list  $(1, 1, 0), (1, 0, 0), (0, 0, 1)$  spans  $\mathbb{R}^3$ :  $(x, y, z) = y(1, 1, 0) + (x - y)(1, 0, 0) + z(0, 0, 1)$ . Stop.

The result is a basis of  $\mathbb{R}^3$  that contains the starting vector.

**Theorem 0.10** (An independent list in a subspace has at most  $\dim$  vectors). *Let  $\mathcal{S}$  be a subspace of  $\mathbb{R}^p$  with  $\dim(\mathcal{S}) = d$  (Definition 0.8).*

1. *Any linearly independent list of vectors in  $\mathcal{S}$  has at most  $d$  vectors.*
2. *If  $\mathcal{T}$  is a subspace with  $\mathcal{T} \subseteq \mathcal{S}$ , then  $\dim(\mathcal{T}) \leq \dim(\mathcal{S})$ .*

**i** Proof

*Proof. Part 1.* If  $d = 0$ , then  $\mathcal{S} = \{\tilde{0}\}$ , so every vector in a nonempty list from  $\mathcal{S}$  is  $\tilde{0}$ , and  $1 \cdot \tilde{0} = \tilde{0}$  shows the list is not linearly independent; only the empty list, with  $0 = d$  vectors, is. Now let  $d \geq 1$ , and let  $\tilde{a}_1, \dots, \tilde{a}_d$  be a basis of  $\mathcal{S}$ , and let  $\tilde{u}_1, \dots, \tilde{u}_k \in \mathcal{S}$  be

linearly independent. Each  $\tilde{u}_j$  is in  $\mathcal{S}$ , so  $\tilde{u}_j = c_{1j}\tilde{a}_1 + \cdots + c_{dj}\tilde{a}_d$  for some coordinate vector  $\tilde{c}_j = (c_{1j}, \dots, c_{dj}) \in \mathbb{R}^d$ . Suppose  $k > d$ . Then  $\tilde{c}_1, \dots, \tilde{c}_k$  are  $k > d$  vectors in  $\mathbb{R}^d$ , so some  $x_1, \dots, x_k$ , not all zero, satisfy  $x_1\tilde{c}_1 + \cdots + x_k\tilde{c}_k = \tilde{0}_d$  (Theorem 0.5); entry  $i$  of that equation says  $\sum_{j=1}^k x_j c_{ij} = 0$ . Then

$$\begin{aligned}
\sum_{j=1}^k x_j \tilde{u}_j &= \sum_{j=1}^k x_j \sum_{i=1}^d c_{ij} \tilde{a}_i && \text{(substitute each } \tilde{u}_j \text{)} \\
&= \sum_{j=1}^k \sum_{i=1}^d x_j c_{ij} \tilde{a}_i && \text{(distribute each } x_j \text{ over the inner sum)} \\
&= \sum_{i=1}^d \sum_{j=1}^k x_j c_{ij} \tilde{a}_i && \text{(swap the order of the finite sums)} \\
&= \sum_{i=1}^d \left( \sum_{j=1}^k x_j c_{ij} \right) \tilde{a}_i && \text{(factor } \tilde{a}_i \text{ out of the inner sum)} \\
&= \sum_{i=1}^d 0 \cdot \tilde{a}_i && \text{(entry } i \text{ of } \sum_j x_j \tilde{c}_j = \tilde{0}_d \text{)} \\
&= \tilde{0}_p, && \text{(a zero multiple of a vector is } \tilde{0} \text{)}
\end{aligned}$$

a combination of  $\tilde{u}_1, \dots, \tilde{u}_k$  with coefficients not all zero that equals  $\tilde{0}_p$ . That contradicts their linear independence, so  $k \leq d$ .

**Part 2.**  $\mathcal{T}$  has a basis (Theorem 0.9) with  $\dim(\mathcal{T})$  vectors. Those vectors are linearly independent and lie in  $\mathcal{S}$ , so by Part 1 there are at most  $\dim(\mathcal{S})$  of them.  $\square$

**Example 0.19** (Subspaces of a plane have dimension at most two). The plane  $\mathcal{S} = \{(x, y, 0) : x, y \in \mathbb{R}\}$  in  $\mathbb{R}^3$  has dimension 2 (Example 0.16). By Theorem 0.10, no three vectors with third entry 0 are linearly independent; for instance,  $(1, 0, 0) + (0, 1, 0) - (1, 1, 0) = \tilde{0}$ . The line  $\mathcal{T} = \text{span}\{(1, 1, 0)\}$  lies in  $\mathcal{S}$ , and

$$\begin{aligned}
\dim(\mathcal{T}) &= 1 \\
&\leq 2 \\
&= \dim(\mathcal{S}).
\end{aligned}$$

**Theorem 0.11** (A subspace inside another of the same dimension equals it). *If  $\mathcal{T}$  and  $\mathcal{S}$  are subspaces of  $\mathbb{R}^p$  with  $\mathcal{T} \subseteq \mathcal{S}$  and  $\dim(\mathcal{T}) = \dim(\mathcal{S})$  (Definition 0.8), then  $\mathcal{T} = \mathcal{S}$ .*

**i** Proof

*Proof.* Let  $d = \dim(\mathcal{S})$ , and let  $\tilde{t}_1, \dots, \tilde{t}_d$  be a basis of  $\mathcal{T}$  (Theorem 0.9); it has  $d$  vectors because  $\dim(\mathcal{T}) = d$ . (When  $d = 0$  it is the empty list, whose span is  $\{\tilde{0}\}$  by the convention in Definition 0.5, and the argument is unchanged.) Suppose some  $\tilde{s} \in \mathcal{S}$  is not in  $\text{span}\{\tilde{t}_1, \dots, \tilde{t}_d\} = \mathcal{T}$ . Then  $\tilde{t}_1, \dots, \tilde{t}_d, \tilde{s}$  are linearly independent (Theorem 0.8), and they are  $d + 1$  vectors of  $\mathcal{S}$ , which contradicts part 1 of Theorem 0.10. So every vector of  $\mathcal{S}$  is in  $\mathcal{T}$ , that is,  $\mathcal{S} \subseteq \mathcal{T}$ . Together with  $\mathcal{T} \subseteq \mathcal{S}$ , this inclusion gives  $\mathcal{T} = \mathcal{S}$ .  $\square$

**Example 0.20** (Two independent vectors in a plane span it). Let  $\mathcal{S} = \{(x, y, 0) : x, y \in \mathbb{R}\}$ , the plane  $z = 0$ , which has dimension 2 (Example 0.16), and let  $\mathcal{T} = \text{span}\{(1, 0, 0), (1, 1, 0)\}$ . Both vectors have third entry 0, so  $\mathcal{T} \subseteq \mathcal{S}$ . They are linearly independent:  $c_1(1, 0, 0) + c_2(1, 1, 0) = (c_1 + c_2, c_2, 0)$ , which is  $\tilde{0}$  only if  $c_2 = 0$  and then  $c_1 = 0$ . So  $\dim(\mathcal{T}) = 2$ , and Theorem 0.11 gives  $\mathcal{T} = \mathcal{S}$  without solving for any coefficients. As a check,  $(x, y, 0) = (x - y)(1, 0, 0) + y(1, 1, 0)$ .

**Example 0.21** (Equal dimensions without containment). The lines  $\text{span}\{(1, 0, 0)\}$  and  $\text{span}\{(0, 1, 0)\}$  in  $\mathbb{R}^3$  both have dimension 1, but they are not equal:  $(1, 0, 0)$  is in the first and not the second. Theorem 0.11 does not apply, because neither line contains the other.

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